

Semi-supervised Knowledge Transfer for Deep Learning from Private Training Data

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Talwar

Aggregation Mechanism

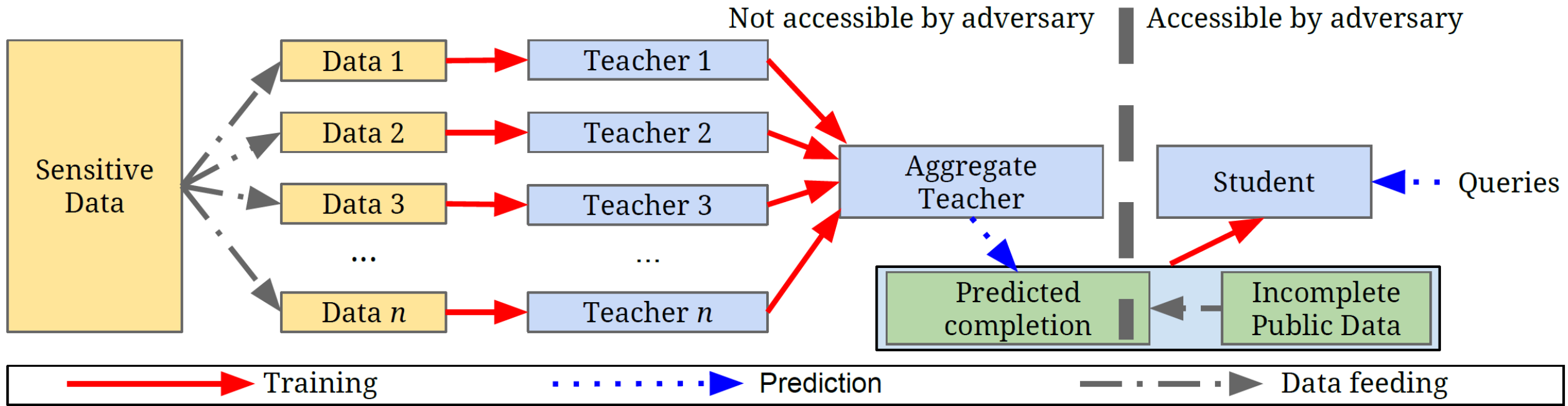
$$n_j(\vec{x}) = |\{i : i \in [n], f_i(\vec{x}) = j\}|.$$

simply apply *plurality*—use the label with the largest count—the ensemble’s decision may depend on a single teacher’s vote. Indeed, when two labels have a vote count differing by at most one, there is a tie: the aggregated output changes if one teacher makes a different prediction. We add random noise to the vote counts n_j to introduce ambiguity:

$$f(x) = \arg \max_j \left\{ n_j(\vec{x}) + \text{Lap} \left(\frac{1}{\gamma} \right) \right\}$$

In this equation, γ is a privacy parameter and $\text{Lap}(b)$ the Laplacian distribution with location 0 and scale b . The parameter γ influences the privacy guarantee we can prove. Intuitively, a large γ leads to a strong privacy guarantee, but can degrade the accuracy of the labels, as the noisy maximum f above can differ from the true plurality.

Teacher-Student



privacy loss to a value that does not grow with the number of user queries made to the student model. Indeed, the privacy loss is now determined by the number of queries made to the teacher ensemble during student training and does not increase as end-users query the deployed student model. Thus

zCDP Properties

Definition 1 (Zero-Concentrated Differential Privacy (zCDP)). A randomised mechanism $M : \mathcal{X}^n \rightarrow \mathcal{Y}$ is (ξ, ρ) -zero-concentrated differentially private (henceforth (ξ, ρ) -zCDP) if, for all $x, x' \in \mathcal{X}^n$ differing on a single entry and all $\alpha \in (1, \infty)$,

$$D_\alpha (M(x) \| M(x')) \leq \xi + \rho\alpha, \quad (1)$$

where $D_\alpha (M(x) \| M(x'))$ is the α -Rényi divergence³ between the distribution of $M(x)$ and the distribution of $M(x')$.

We define ρ -zCDP to be $(0, \rho)$ -zCDP.⁴

Definition 14 (Rényi Divergence [Ren61, Eq. (3.3)]). Let P and Q be probability distributions on Ω . For $\alpha \in (1, \infty)$, we define the Rényi divergence of order α between P and Q as

$$\begin{aligned} D_\alpha (P \| Q) &= \frac{1}{\alpha - 1} \log \left(\int_{\Omega} P(x)^\alpha Q(x)^{1-\alpha} dx \right) \\ &= \frac{1}{\alpha - 1} \log \left(\mathbb{E}_{x \sim P} \left[\left(\frac{P(x)}{Q(x)} \right)^{\alpha-1} \right] \right), \end{aligned}$$

Proposition 3. If M provides ρ -zCDP, then M is $(\rho + 2\sqrt{\rho \log(1/\delta)}, \delta)$ -differentially private for any $\delta > 0$.

Proposition 4. If M satisfies ε -differential privacy, then M satisfies $(\frac{1}{2}\varepsilon^2)$ -zCDP.

Definition 2. Let $\mathcal{M}: \mathcal{D} \rightarrow \mathcal{R}$ be a randomized mechanism and d, d' a pair of adjacent databases. Let $\mathbf{aux}$ denote an auxiliary input. For an outcome $o \in \mathcal{R}$, the privacy loss at o is defined as:

$$c(o; \mathcal{M}, \mathbf{aux}, d, d') \triangleq \log \frac{\Pr[\mathcal{M}(\mathbf{aux}, d) = o]}{\Pr[\mathcal{M}(\mathbf{aux}, d') = o]}. \quad (3)$$

$C(\mathcal{M}, \mathbf{aux}, d, d')$ is defined as $c(\mathcal{M}(d); \mathcal{M}, \mathbf{aux}, d, d')$

Definition 3. Let $\mathcal{M}: \mathcal{D} \rightarrow \mathcal{R}$ be a randomized mechanism and d, d' a pair of adjacent databases. Let $\mathbf{aux}$ denote an auxiliary input. The moments accountant is defined as:

$$\alpha_{\mathcal{M}}(\lambda) \triangleq \max_{\mathbf{aux}, d, d'} \alpha_{\mathcal{M}}(\lambda; \mathbf{aux}, d, d') \quad (4)$$

$$\alpha_{\mathcal{M}}(\lambda; \mathbf{aux}, d, d') \triangleq \log \mathbb{E}[\exp(\lambda C(\mathcal{M}, \mathbf{aux}, d, d'))]$$

Basic Notation

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$$= \log \sum_{o \in \mathcal{R}(d)} \left[\exp(\lambda \cdot \log \frac{\Pr[\mathcal{M}(d)=o]}{\Pr[\mathcal{M}(d')=o]}) \right] = \lambda \cdot \frac{1}{\lambda} \cdot \log \mathbb{E} \left(\frac{\Pr[\mathcal{M}(d)=o]}{\Pr[\mathcal{M}(d')=o]} \right)^\lambda$$

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$$= \lambda \cdot \frac{1}{\lambda} \log \mathbb{E}_{o \sim M(d)} \left[\frac{P_{\mathcal{M}}(M(d) \leq o)}{P_{\mathcal{M}}(M(d) \leq 0)} \right] = \lambda \cdot D_{\lambda+1}(M(d) \| M(d'))$$

Important Properties

Theorem 1. 1. *[Composability] Suppose that a mechanism $\mathcal{M}$ consists of a sequence of adaptive mechanisms $\mathcal{M}_1, \dots, \mathcal{M}_k$ where $\mathcal{M}_i: \prod_{j=1}^{i-1} \mathcal{R}_j \times \mathcal{D} \rightarrow \mathcal{R}_i$. Then, for any output sequence $o_1, \dots, o_{k-1}$ and any λ*

$$\alpha_{\mathcal{M}}(\lambda; d, d') = \sum_{i=1}^k \alpha_{\mathcal{M}_i}(\lambda; o_1, \dots, o_{i-1}, d, d') ,$$

where $\alpha_{\mathcal{M}}$ is conditioned on $\mathcal{M}_i$'s output being o_i for $i < k$.

2. *[Tail bound] For any $\varepsilon > 0$, the mechanism $\mathcal{M}$ is (ε, δ) -differentially private for*

$$\delta = \min_{\lambda} \exp(\alpha_{\mathcal{M}}(\lambda) - \lambda\varepsilon) .$$

Privacy Loss Analysis

Theorem 2. *Suppose that on neighboring databases d, d' , the label counts n_j differ by at most l in each coordinate. Let $\mathcal{M}$ be the mechanism that reports $\arg \max_j \left\{ n_j + \text{Lap}(\frac{1}{\gamma}) \right\}$. Then $\mathcal{M}$ satisfies $(2\gamma, 0)$ -differential privacy. Moreover, for any l , $\mathbf{aux}$, d and d' ,*

$$\alpha(l; \mathbf{aux}, d, d') \leq 2\gamma^2 l(l + 1) \tag{5}$$

T steps, we get $(4T\gamma^2 + 2\gamma\sqrt{2T \ln \frac{1}{\delta}}, \delta)$ -differential privacy.

Theorem 2. Suppose that on neighboring databases d, d' , the label counts n_j differ by at most 1 in each coordinate. Let $\mathcal{M}$ be the mechanism that reports $\arg \max_j \left\{ n_j + \text{Lap}\left(\frac{1}{\gamma}\right) \right\}$. Then $\mathcal{M}$ satisfies $(2\gamma, 0)$ -differential privacy. Moreover, for any l , $\mathbf{aux}$, d and d' ,

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$\Rightarrow M$ satisfies $2\gamma^2$ -zCDP $\leftarrow$

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Proposition 4. If M satisfies ε -differential privacy, then M satisfies $(\frac{1}{2}\varepsilon^2)$ -zCDP.

recall $\alpha(l; \mathbf{aux}, d, d') = 1 - \mathbb{P}_{\mathcal{M}}(M(d) \neq M(d'))$

we have it
 $\leq 6 \cdot 2\gamma^2 \cdot (l+1)$

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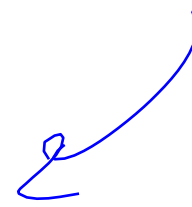
$$\alpha_{\mathcal{M}}(\lambda; d, d') = \sum_{i=1}^k \alpha_{\mathcal{M}_i}(\lambda; o_1, \dots, o_{i-1}, d, d'),$$

where $\alpha_{\mathcal{M}}$ is conditioned on $\mathcal{M}_i$'s output being o_i for $i < k$.

$$\leq T \cdot 2\gamma^2 (L+1)$$

$$\Rightarrow \rho = 2\gamma^2 T$$

set $k=T$



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where $\alpha_{\mathcal{M}}$ is conditioned on $\mathcal{M}_i$'s output being o_i for $i < k$.

$$\rho = 2\gamma^2 T \Rightarrow \underline{4T\gamma^2} + 2\gamma\sqrt{2T \cdot \log(\frac{1}{\delta})} \cdot \delta - dp$$

Proposition 3. If M provides ρ -zCDP, then M is $(\rho + 2\sqrt{\rho \log(1/\delta)}, \delta)$ -differentially private for any $\delta > 0$.

Our data-dependent privacy analysis takes advantage of the fact that when the quorum among the teachers is very strong, the majority outcome has overwhelming likelihood, in which case the privacy cost is small whenever this outcome occurs. The moments accountant allows us analyze the composition of such mechanisms in a unified framework.

Theorem 3. Let $\mathcal{M}$ be $(2\gamma, 0)$ -differentially private and $q \geq \Pr[\mathcal{M}(d) \neq o^*]$ for some outcome o^* . Let $l, \gamma \geq 0$ and $q < \frac{e^{2\gamma} - 1}{e^{4\gamma} - 1}$. Then for any $\mathbf{aux}$ and any neighbor d' of d , $\bar{\mathcal{M}}$ satisfies

when o is
high q is
small

$$\alpha(l; \mathbf{aux}, d, d') \leq \log\left((1 - q) \left(\frac{1 - q}{1 - e^{2\gamma}q}\right)^l + q \exp(2\gamma l)\right).$$

$(0, e^{-2\gamma})$: increasing

assumed
overwhelming
class

Lemma 4. Let $\mathbf{n}$ be the label score vector for a database d with $n_{j^*} \geq n_j$ for all j . Then

$$\Pr[\mathcal{M}(d) \neq j^*] \leq \sum_{j \neq j^*} \frac{2 + \gamma(n_{j^*} - n_j)}{4 \exp(\gamma(n_{j^*} - n_j))}$$

same as
of

$$= \sum_{i \neq i^*} \Pr(M(d) = o_i) \geq e^{2\gamma} \sum_{i \neq i^*} \Pr(M(d') = o_i) = e^{2\gamma} q''$$

$$q'' \leq e^{2\gamma} q'$$

Proof. Since $\mathcal{M}$ is 2γ -differentially private, for every outcome o , $\frac{\Pr[M(d)=o]}{\Pr[M(d')=o]} \leq \exp(2\gamma)$. Let $q' = \Pr[M(d) \neq o^*]$. Then $\Pr[M(d') \neq o^*] \leq \exp(2\gamma)q'$. Thus

$$\begin{aligned} \exp(\alpha(l; \mathbf{aux}, d, d')) &= \sum_o \Pr[M(d) = o] \left(\frac{\Pr[M(d) = o]}{\Pr[M(d') = o]} \right)^l \\ &= \underbrace{\Pr[M(d) = o^*]}_{\text{same as } \Pr[M(d') = o^*]} \left(\frac{\Pr[M(d) = o^*]}{\Pr[M(d') = o^*]} \right)^l + \sum_{o \neq o^*} \Pr[M(d) = o] \left(\frac{\Pr[M(d) = o]}{\Pr[M(d') = o]} \right)^l \\ &\leq (1 - q') \left(\frac{1 - q'}{1 - e^{2\gamma}q'} \right)^l + \sum_{o \neq o^*} \Pr[M(d) = o] (e^{2\gamma})^l \\ &\leq (1 - q') \left(\frac{1 - q'}{1 - e^{2\gamma}q'} \right)^l + q' e^{2\gamma l}. \end{aligned}$$

Now consider the function

$$f(z) = (1 - z) \left(\frac{1 - z}{1 - e^{2\gamma} z} \right)^l + z e^{2\gamma l}.$$

We next argue that this function is non-decreasing in $(0, \frac{e^{2\gamma}-1}{e^{4\gamma}-1})$ under the conditions of the lemma. Towards this goal, define

$$g(z, w) = (1 - z) \left(\frac{1 - w}{1 - e^{2\gamma} w} \right)^l + z e^{2\gamma l},$$

and observe that $f(z) = g(z, z)$. We can easily verify by differentiation that $g(z, w)$ is increasing individually in z and in w in the range of interest. This implies that $f(q') \leq f(q)$ completing the proof. $\square$

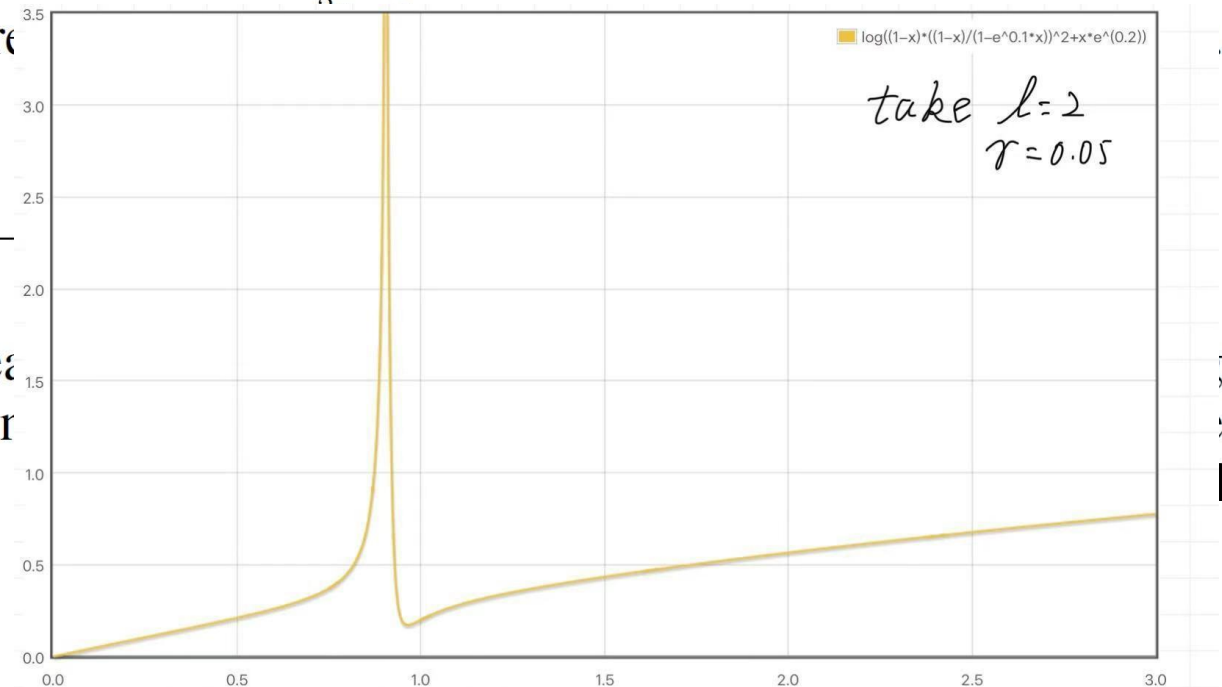
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We next argue that this function is non-decreasing.
Towards this goal, define

$$g(z, w) = (1 -$$

and observe that $f(z) = g(z, z)$. We can establish the result by showing that g is non-decreasing individually in z and in w in the range of interest for the proof.



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$$\begin{aligned} \frac{\partial g}{\partial z} &= e^{2\gamma l} - \left(\frac{1-w}{1-e^{2\gamma} w} \right)^l \\ \frac{\partial g}{\partial w} &= \left(\frac{e^{2\gamma} - e^{4\gamma} w - 1 + w}{1 - e^{2\gamma} w} \right)^l \end{aligned}$$

$$\begin{aligned} &\underline{e^{2\gamma} - 1 + (1 - e^{4\gamma} w)} = t(w) \\ \text{when } t(w) \geq 0 : & e^{2\gamma} - 1 + (1 - e^{4\gamma} w) \geq 0 \end{aligned}$$

$$w \leq \frac{e^{2\gamma} - 1}{e^{4\gamma} - 1}$$

bigger than $\frac{e^{2\gamma} - 1}{e^{4\gamma} - 1}$

We next argue that this function is non-decreasing in $(0, \frac{e^{2\gamma}-1}{e^{4\gamma}-1})$ under the conditions of the lemma. Towards this goal, define

$$g(z, w) = \underbrace{(1-z)}_{>0} \left(\frac{1-w}{1-e^{2\gamma}w} \right)^{\underbrace{l}_{>1}} + ze^{2\gamma l},$$

and observe that $f(z) = g(z, z)$. We can easily verify by differentiation that $g(z, w)$ is increasing individually in z and in w in the range of interest. This implies that $f(q') \leq f(q)$ completing the proof. $\square$

$$h(w) = \frac{1-w}{1-e^{2\gamma}w}$$

$$\frac{\partial g}{\partial w} \text{ have same sign as } \frac{\partial h(w)}{\partial w} = \frac{-(1-e^{2\gamma}w) - (1-w)(e^{2\gamma})}{(1-e^{2\gamma}w)^2}$$

$$\frac{\partial g}{\partial w} = \frac{e^{2\gamma}w - 1 + e^{2\gamma} - e^{2\gamma}w}{(1-e^{2\gamma}w)^2}$$

$$= \frac{e^{2\gamma} - 1}{(1-e^{2\gamma}w)^2} > 0$$

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and observe that $f(z) = g(z, z)$. We can easily verify by differentiation that $g(z, w)$ is increasing individually in z and in w in the range of interest. This implies that $f(q') \leq f(q)$ completing the proof. $\square$

$\gamma = 0.05, T = 1000, \delta = 1\text{e-}6$ gives us $\varepsilon \approx 26$

SVHN	8.19	10^{-6}	1000	92.80%	90.66%
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	Dataset	ε	δ	Queries	Non-Private Baseline	Student Accuracy	
We notice that the accuracy of the noisy aggregation is increasing as the number of teachers increases. Toward the non-private baseline.	MNIST	2.04	10^{-5}	100	99.18%	98.00%	a.
	MNIST	8.03	10^{-5}	1000	99.18%	98.10%	
	SVHN	5.04	10^{-6}	500	92.80%	82.72%	
	SVHN	8.19	10^{-6}	1000	92.80%	90.66%	

and observe that $acc(\gamma) = acc(\delta, \gamma)$. We can easily verify by differentiation that $acc(\delta, \gamma)$ is increasing in γ . This implies that the accuracy of the noisy aggregation is increasing as the number of teachers increases. Toward the non-private baseline.

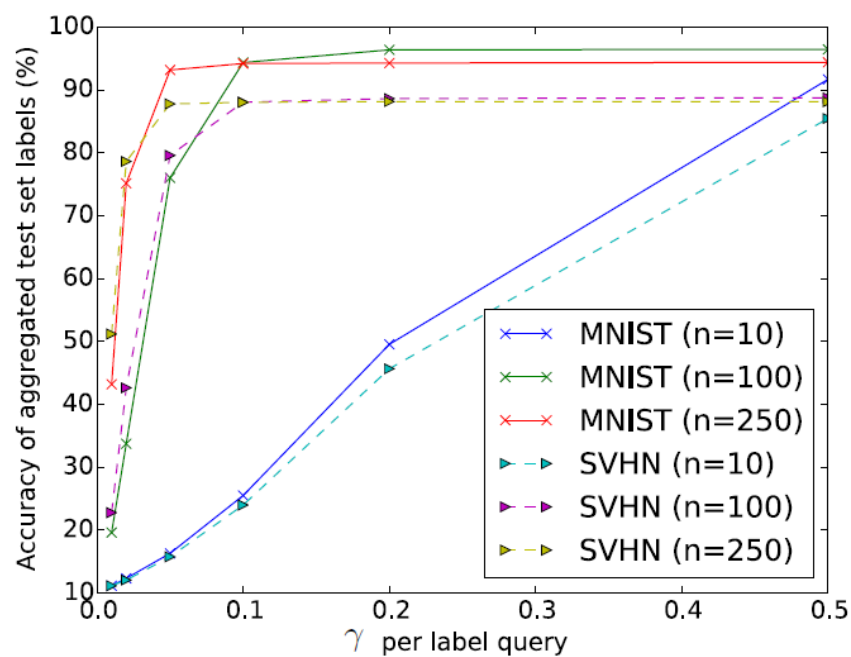


Figure 2: **How much noise can be injected to a query?** Accuracy of the noisy aggrega-

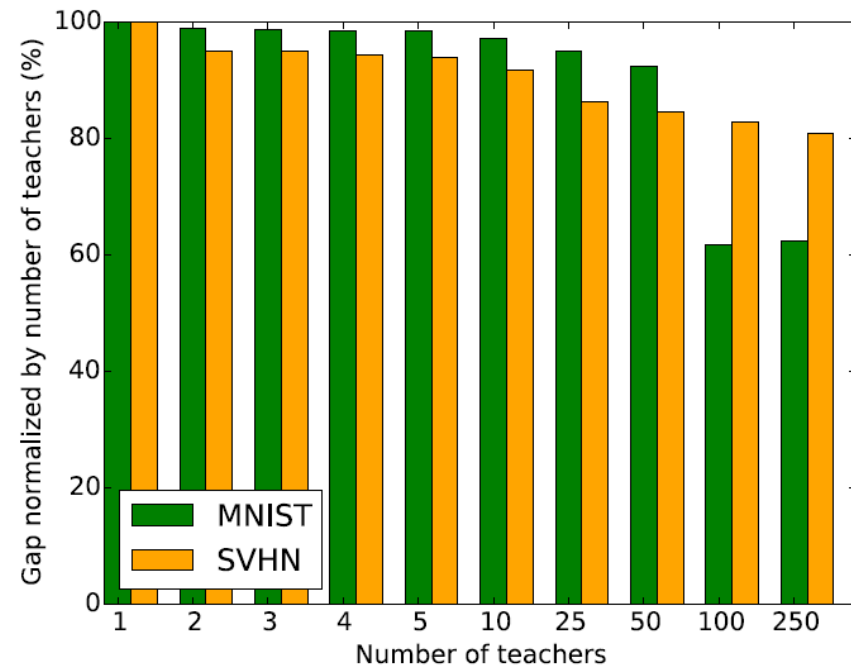


Figure 3: **How certain is the aggregation of teacher predictions?** Gap between the num-